

Féidearthachtaí as Cuimse
Infinite Possibilities

Management Accounting

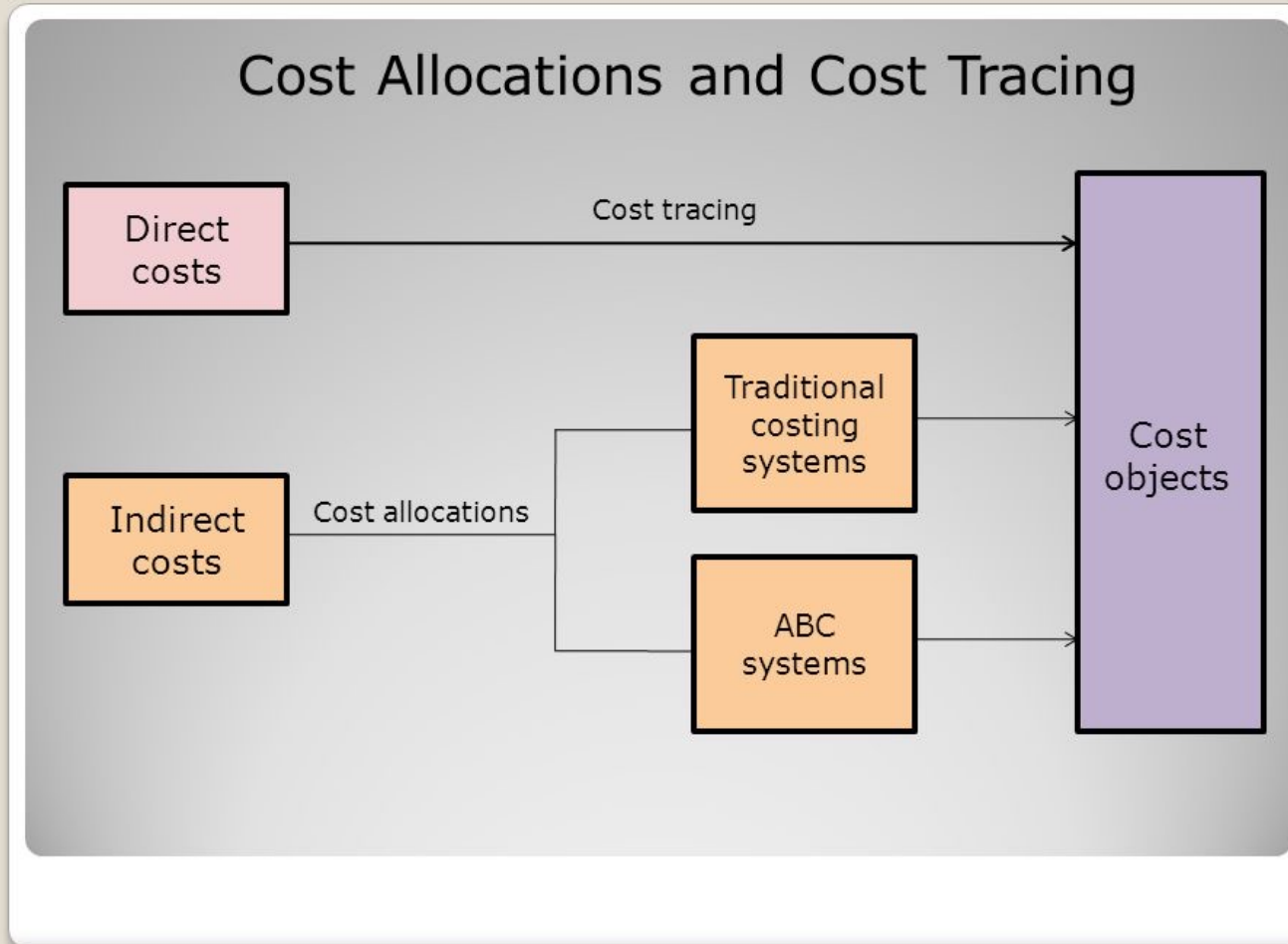
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Lecture 9: Overhead Allocation - ABC

Semester 1 - 2023/24



Recap: Tracing Direct/Indirect Costs



Direct materials
Direct labour
Direct expenses
Direct cost
Production overhead
Full production cost
Non-production cost
Total cost

Recap: Overhead absorption method

- 1) Allocate production overheads to production and service cost centres.
- 2) Reapportion service centre costs to production centres.
- 3) Calculate Overhead Absorption Rates (OARs) for each production centre.
- 4) Use OARs to cost products.

Recap: Step 1) Allocate production overheads

Production Overhead	Typical Bases of Allocation
Factory Rent & Rates	Floor area
Production supervisor's salary	Number of staff or labour hours
Depreciation of machinery	Machine value
Canteen costs	Number of staff
Machinery /Plant /Equipment Insurance	Machine /Plant /Equipment value
Factory Power	Power usage / Machine Horse Power
Light & Heat	Floor area
Personnel/Supervision Costs	Number of Employees
Canteen Costs	Number of Employees

Budgeted or actual OARs?

- Budgeted or predetermined information is used when calculating the OAR.
- Actual information on overheads may not be known until the year end.
- Not all overhead costs may be captured if an OAR is calculated on a regular basis using actual information.
- The problem can be overcome by using budgeted information to calculate the OAR.
- This can result in under-absorption or over-absorption of overheads.

Under-absorbed overheads

- The amount of overhead actually incurred in the year was greater than the amount of overhead absorbed into jobs.
- Accounting treatment: treat the amount of under-absorbed overhead as a period cost and subtract it from profit in the income statement.

	Polishing €
Overhead absorbed: OAR x actual labour hours €8 x 12 000	96 000
Actual overhead	<u>(100 000)</u>
Under-absorbed overhead	<u>(4 000)</u>

Over-absorbed overhead

- The amount of overhead incurred in the year was less than the amount of overhead absorbed into jobs for the year.
- Accounting treatment: add it as miscellaneous revenue to profit in the income statement.

	Finishing €
Overhead absorbed: OAR x actual labour hours €5 x 2 500	12 500
Actual overhead	<u>(8 000)</u>
Over-absorbed overhead	<u>4 500</u>

Activity-based Costing (ABC)

- Traditional methods do not provide a reliable measure of the non-manufacturing overheads consumed by products.
- ABC is advocated for providing a more accurate measure of resources consumed by products.
- Have a greater number of cost centres in the first stage
- have greater number and variety of cost drivers (allocation bases) in the second stage

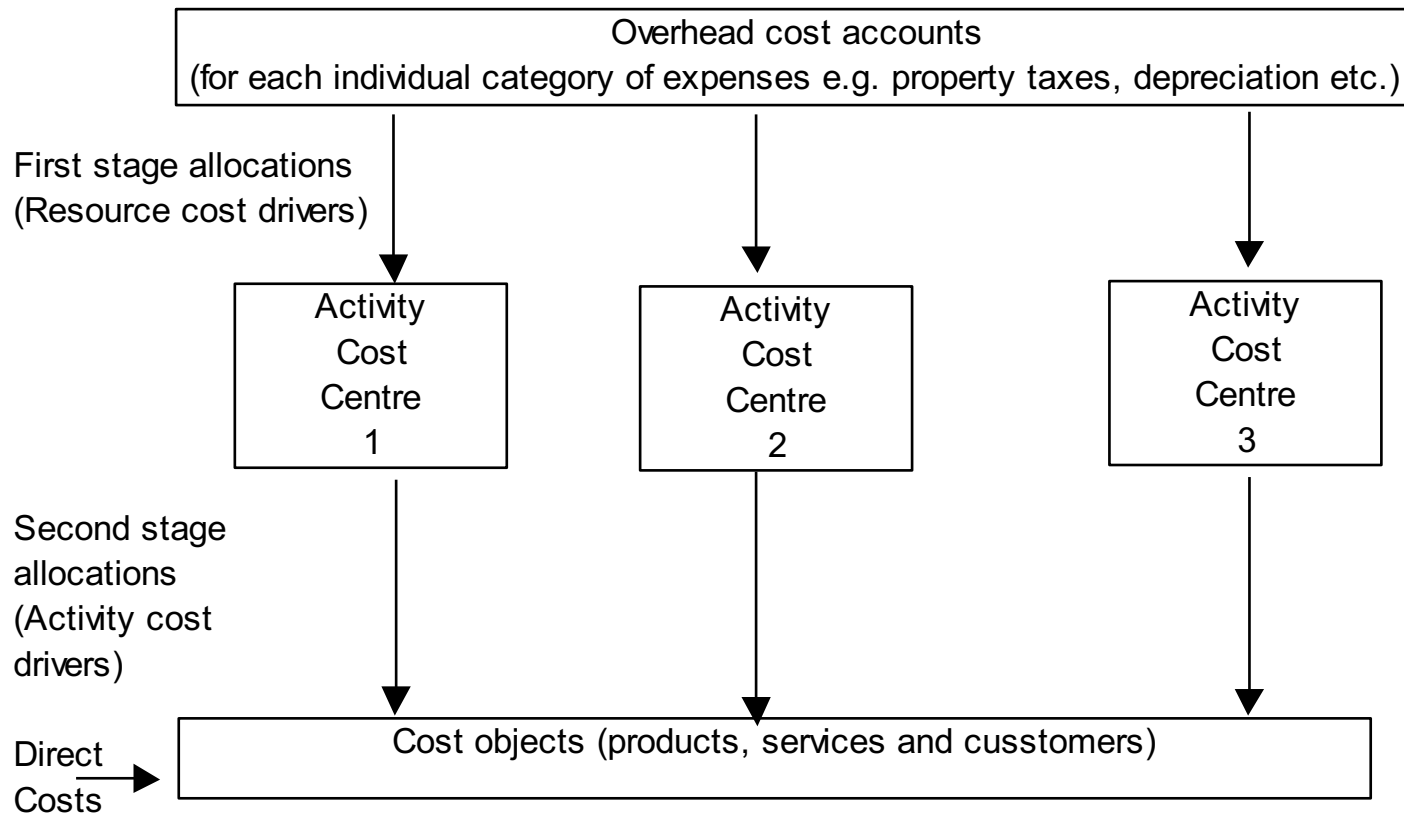
A comparison of traditional and ABC systems

- Both systems use the two-stage allocation process.
- In the first stage traditional systems tend to allocate costs to departments whereas ABC systems allocate costs to activities:
(ABC systems tend to have more cost centres/cost pools)
- In the second stage traditional systems rely on a small number of volume-based cost drivers (typically direct labour or machine hours) whereas ABC systems use many second-stage cost drivers.
- ABC systems seek to use only cause-and-effect cost drivers whereas traditional systems often rely on arbitrary allocation bases.
- ABC systems tend to establish separate cost driver rates for support departments whereas traditional systems merge support (service) and production centre costs.

ABC steps

- Overheads allocated to each major activity (rather than Depts)
- Activities the aggregation of many different tasks and are described by verbs associated with objects eg purchase materials, process supplier records

(b) Activity-based costing systems



1. Identify the major activities that take place in an organization
2. Assign costs to cost pools /cost centre for each activity:
3. Determine the cost driver for each major activity
4. Assign the cost of activities to products

Practice 1

Yang Manufacturing Company manufactures two products (A and B). The overhead costs (€58,000) have been divided into three cost pools that use the following activity cost drivers/allocation bases:

Using **traditional-based costing**, what is the amount of overhead cost to be assigned to Product B using labour hours as the allocation base?

<u>Product</u>	<u>Number of Orders</u>	<u>Number of Labour Transactions</u>	<u>Labour Hours</u>
A	15	50	500
B	10	150	2,000
Cost per pool	€10,000	€8,000	€40,000

- a. €58,000
- b. €43,500
- c. €11,600
- d. €46,400

$$58000 / (200 + 500) * 2000 = 46400$$

Practice 2

Zipp Company manufactures two products (X and Y). The overhead costs (£84,000) have been divided into three cost pools that use the following activity cost drivers:

<u>Product</u>	<u>Number of Setups</u>	<u>Machine Hours</u>	<u>Packing Orders</u>
X	10	500	75
Y	10	2,000	175
Cost per pool	£9,000	£60,000	£15,000

Refer to Figure above. What is the allocation rate per packing order using activity-based costing?

$$15000/(75+175) = 60$$

How much packing order cost based on ABC should be allocated to X?

$$60 * 75 = 4500$$

Practice 3

Q1. A company currently uses an overhead absorption rate based on direct labour hours. The following costings have been prepared for its three products:

Per unit	A135 €	B160€	C243€
Prime costs	3.75	4.89	11.450
Overhead costs	2.00	1.00	2.466
Total Costs	5.75	5.89	13.916
Selling price	10.00	12.00	16.000
Profit / Loss	4.25	6.11	2.084

The company is considering implementing an activity-based costing system. The following information has been collected:

Overheads / activities	Activity cost pools €	Cost Drivers
Machine set ups	2,996	Number of machine set ups
Machinery costs	8,985	Number of machine hours
Quality checks	2,100	Number of inspections
Order processing	885	Number of orders processed
Total	14,996	

And



	A135	B160	C243	Total
Number of machine setups	500	625	373	1,498
Number of machine hours	1,245	1,250	500	2,995
Number of inspections	20	35	15	70
Number of orders processed	2,200	1,130	210	3,540
Units produced and sold	5,000	2,500	1,000	8,500

- Calculate the total unit cost to two decimal places and the profit for the three products using activity-based costing
- Comment on your findings.

Sometimes the cost drivers are not provided, so you need to be able to identify the appropriate cost driver

Practice 4

Q2. O'Brien Ltd manufactures shoes and produces three styles – the sandal, the runner and the boot. They have recently implemented an activity-based costing system. The following data has been collected.

Overheads / Activities	Activity Cost Pools €
Sales and Marketing costs	67,500
Customer Order Costs	273,125
Inspection costs	420,000
Machining costs	860,000
Total	1,620,625

	Sandal	Runner	Boot	Total
Prime cost (per unit)	€35.00	€49.00	€77.00	
Units produced	25,000	15,000	12,500	52,500
Number of sales calls	1,000	1,500	500	3,000
Number of customer orders	12,500	20,000	25,000	57,500
Machine hours	25,000	15,000	10,000	50,000

Required:

- Prepare a statement showing the total overhead cost of products, sandals, runners and boots. Show the overhead cost per unit and the total cost per unit (to two decimal places).
- Calculate the selling price per unit (to two decimal places) for each of the three products on the basis of
 - 10% markup on the cost
 - 20% margin on the selling price

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Any Questions?
Thank you for your participation

